Quantum compilation and resource estimation

(I) Quantum compilation overview

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AlgoLab Summer School – Monday 15 June 2026

THE UNIVERSITY OF BRITISH COLUMBIA
Electrical and Computer Engineering
Faculty of Applied Science

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QUANTUM SOFTWARE & ALGORITHMS RESEARCH

Stewart Blusson Quantum Matter Institute
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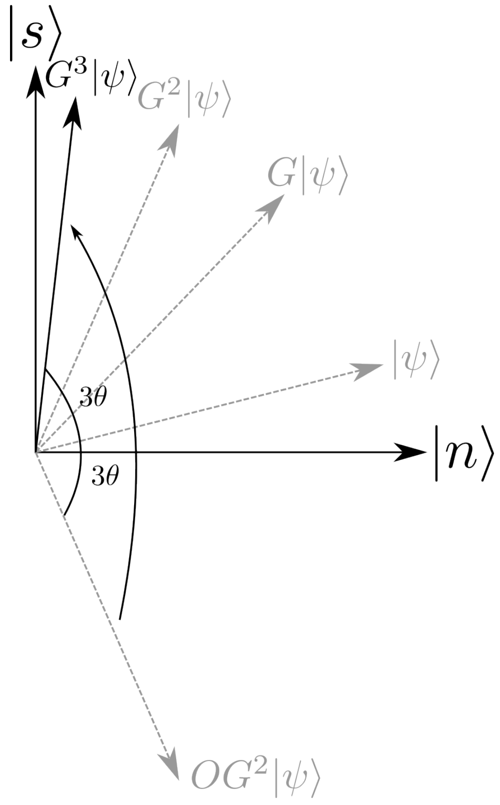
Expectations vs. reality

“Quantum computers will be able to solve every problem exponentially faster.”

Gotta go fast!

Grover's algorithm is one of the few quantum algorithms with a **provable quantum speed up.**

Oracle queries to find a marked item in an *N*-bit search space:



Question

How much does it cost to run a preimage attack on the **SHA2-256** hash function with Grover search on a fault-tolerant quantum computer?

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How much does it cost to run a preimage attack on the **SHA2-256** hash function with Grover search on a fault-tolerant quantum computer?

- (A) 2^{128}
- (B) 2^{145}
- (C) 2^{166}
- (D) $2^{127.6}$

Resource estimation is *important*

arXiv > quant-ph > arXiv:2505.15917

Search...

Help | Adv

Quantum Physics

[Submitted on 21 May 2025]

How to factor 2048 bit RSA integers with less than a million noisy qubits

Craig Gidney

GIZNEWS DAILY

One-Million-Qubit Machine Can Break 2048-Bit RSA in 7 Days

Written by GizNews in All News, Computers May 27, 2025

arXiv > quant-ph > arXiv:2603.28627

Search...

Help | A

Quantum Physics

[Submitted on 30 Mar 2026]

Shor's algorithm is possible with as few as 10,000 reconfigurable atomic qubits

Madelyn Cain, Qian Xu, Robbie King, Lewis R. B. Picard, Harry Levine, Manuel Endres, John Preskill, Hsin-Yuan Huang, Dolev Bluvstein

ScienceNews

NEWS QUANTUM PHYSICS

Just 10,000 quantum bits might crack internet encryption schemes

Computers based on atoms could demand a fraction of the qubits previously estimated

PHYSICS & CHEMISTRY

How Should We Prepare for the Looming Quantum Encryption Apocalypse?

The dreaded Q-day could arrive sooner than expected, and when it does, experts say we need to be ready.

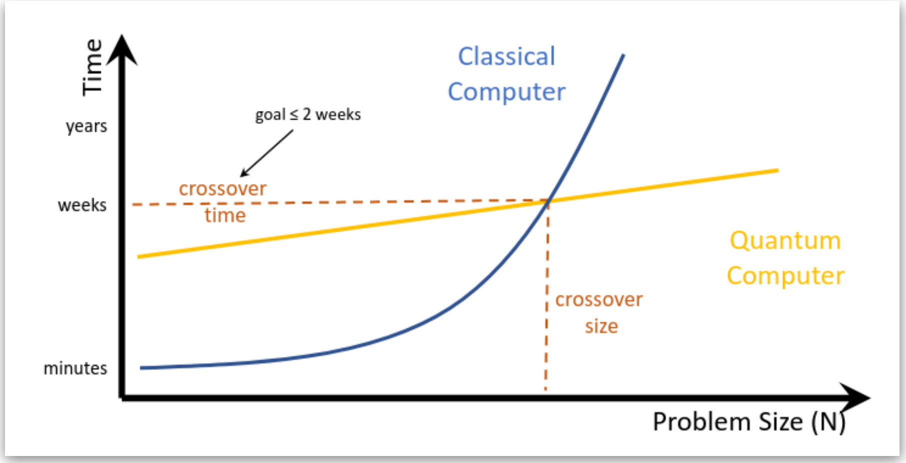
BY GAYOUNG LEE

PUBLISHED APRIL 11, 2026, 7:00 AM ET

Quantum resource estimation

How much does it *really* cost to run a quantum algorithm?

- Qubits?
- Gates?
- Time?
- Energy?
- Money?



Speed vs. scaling

How to answer these questions?

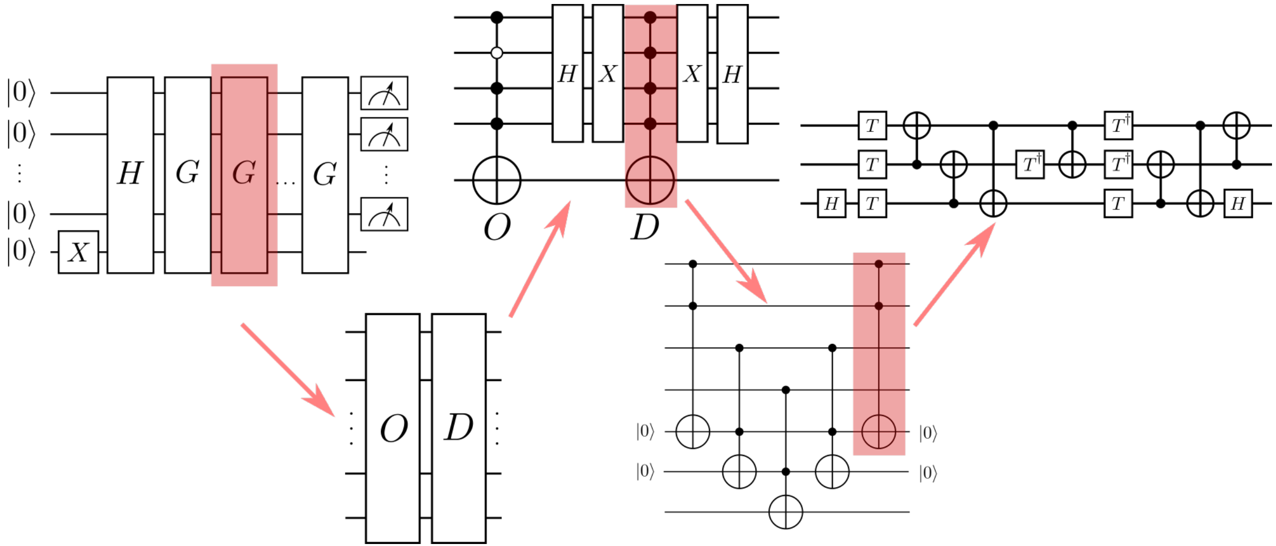
T. Hoefler, T. Häner, and M. Troyer (2023) Disentangling Hype from Practicality: On Realistically Achieving Quantum Advantage. arXiv:2307.00523 [quant-ph].

Quantum compilation and resource estimation

Learning outcomes:

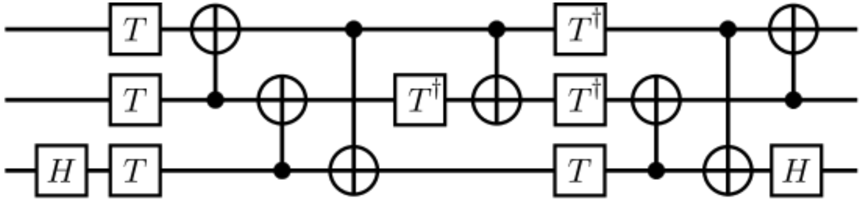
1. List and describe the steps of the quantum compilation pipeline
2. Apply compilation techniques in software and perform logical-level resource estimates
3. Estimate the physical resources required to run an algorithm with error correction

Lecture 1 in one slide



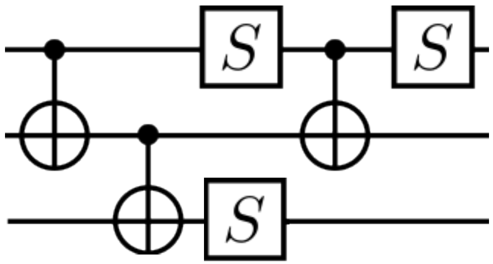
How do we do this in the best way possible? What defines “best”?

What are the relevant **costs**?



Circuit depth

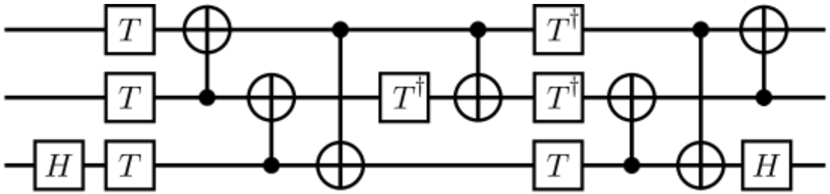
The circuit depth is the length of the longest path in its **directed acyclic graph (DAG)** representation.

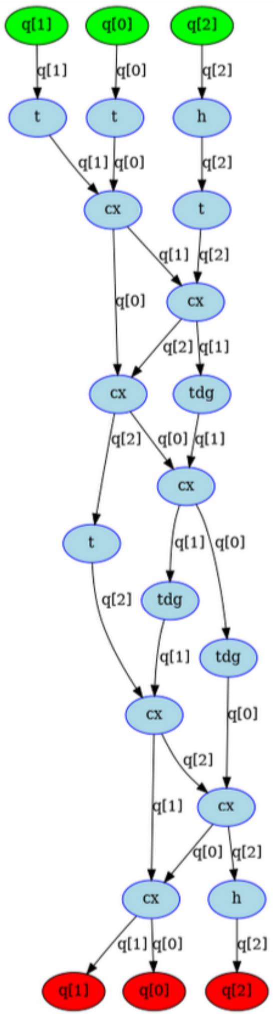


Circuit depth

Generally don’t do by hand;
NetworkX is your friend.

https://networkx.org/

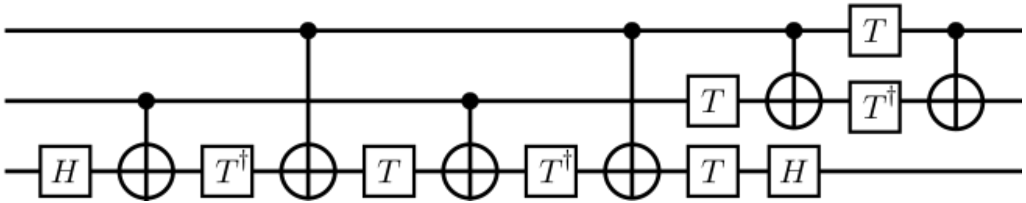




(DAG generated
in Qiskit)

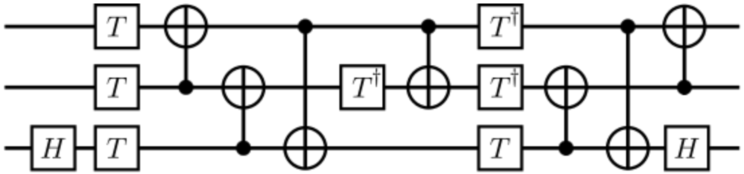
Exercise

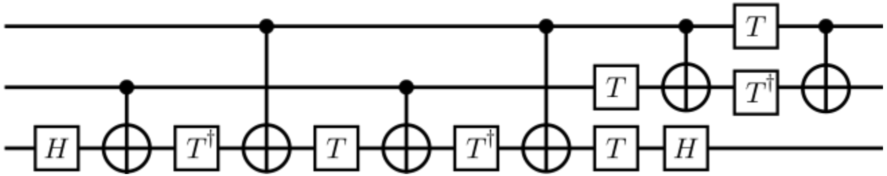
Determine depth, width, and gate counts of the following circuit.



Question

What do these circuits do?





US

High-level algorithmic description

Software
(Human-readable code)

CPU

Compilers are essential

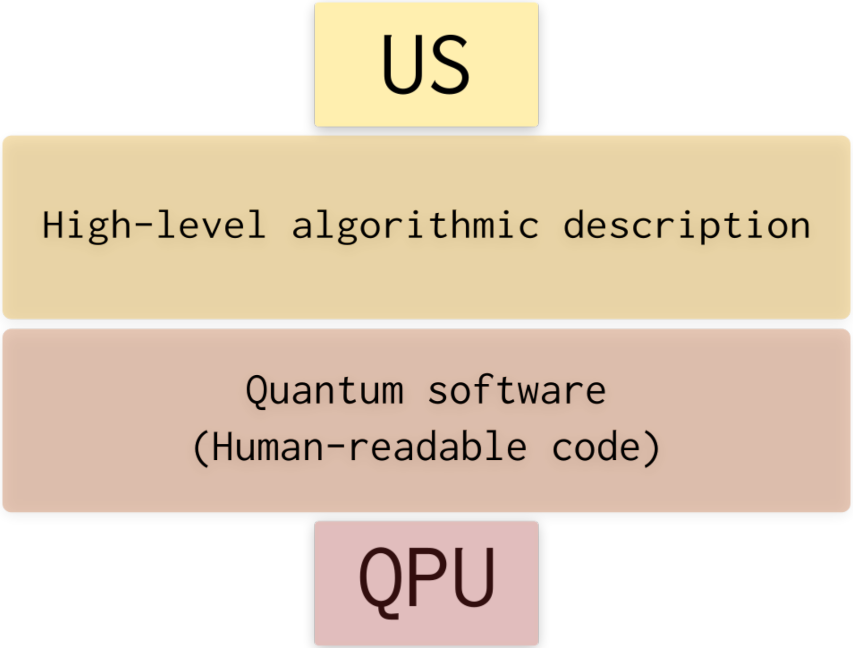
```c
#include<stdio.h>

void main() {
    printf("Hello, world!");
}
```

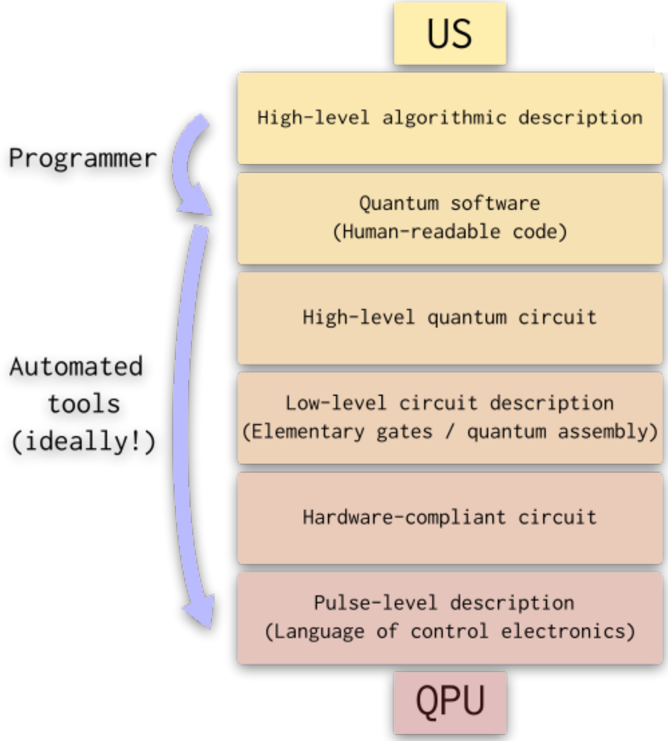
```
        .file   "hello_world.c"
        .text
        .section        .rodata
.LC0:
        .string "Hello, world!"
        .text
        .globl  main
        .type   main, @function
main:
.LFB0:
        .cfi_startproc
        endbr64
        pushq   %rbp
        .cfi_def_cfa_offset 16
        .cfi_offset 6, -16
        movq    %rsp, %rbp
        .cfi_def_cfa_register 6
        leaq    .LC0(%rip), %rdi
        movl    $0, %eax
        call    printf@PLT
        nop
        popq    %rbp
        .cfi_def_cfa 7, 8
        ret
        .cfi_endproc
.LFE0:
        .size   main, .-main
        .ident  "GCC: (Ubuntu 9.2.1-9ubuntu2) 9
```

```
 11a0 ff48c1fd 03741f31 db0f1f80 00000000  .H...t.1........
 11b0 4c89f24c 89ee4489 e741ff14 df4883c3  L..L..D..A...H..
 11c0 014839dd 75ea4883 c4085b5d 415c415d  .H9.u.H...[]A\A]
 11d0 415e415f c366662e 0f1f8400 00000000  A^A_.ff.........
 11e0 f30f1efa c3                          .....
Contents of section .fini:
 11e8 f30f1efa 4883ec08 4883c408 c3        ....H...H....
Contents of section .rodata:
 2000 01000200 48656c6c 6f2c2077 6f726c64  ....Hello, world
 2010 2100                                 !.
Contents of section .eh_frame_hdr:
 2014 011b033b 40000000 07000000 0cf0ffff  ...;@...........
 2024 74000000 2cf0ffff 9c000000 3cf0ffff  t...,.......<...
 2034 b4000000 4cf0ffff 5c000000 35f1ffff  ....L...\...5...
 2044 cc000000 5cf1ffff ec000000 ccf1ffff  ....\...........
 2054 34010000                             4...
Contents of section .eh_frame:
```

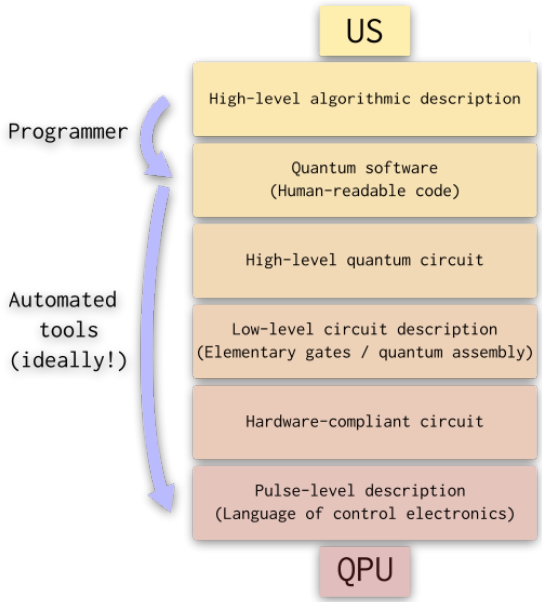
In an ideal world…

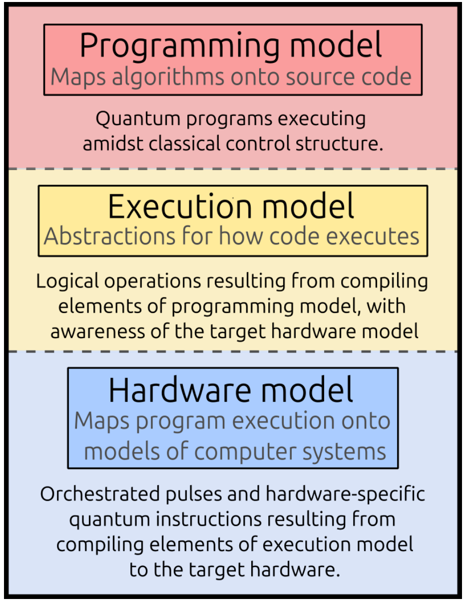


High-level overview: quantum compilation stack



Quantum compilation and abstraction

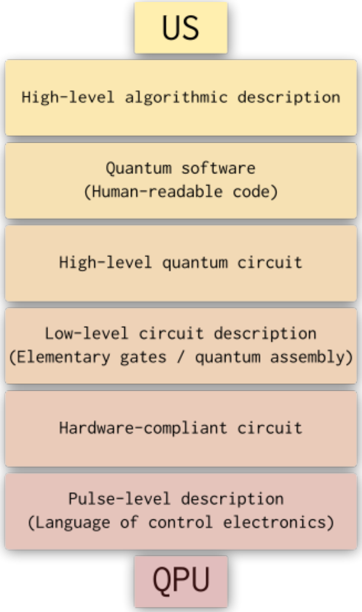




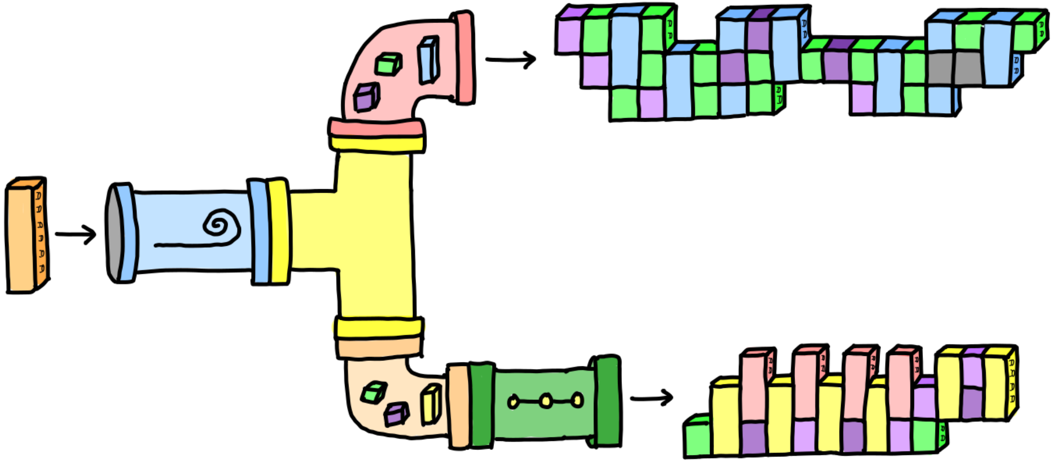
ODM, S. Núñez-Corrales, M. Stęchły, S. Reinhardt, and T. Mattson. 2024. An abstraction hierarchy towards productive quantum programming. In Proc. of 2024 IEEE International Conference on Quantum Computing and Engineering (QCE), Montreal, PQ. pp. 979-989.

*Ask me later about more
general program compilation

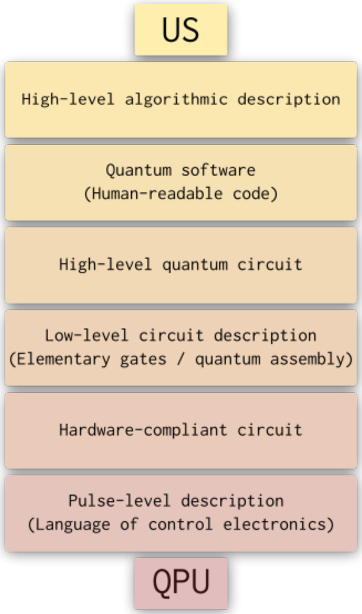
Quantum circuit* compilation



Implemented as a sequence of modular ***passes***, or ***transforms***, that modify a circuit in various ways.



Quantum circuit compilation



Implemented as a sequence of modular ***passes***, or ***transforms***, that modify a circuit in various ways.

Loosely divided into 3 stages:

- decomposition and synthesis
- hardware-independent optimization
- hardware-dependent optimization

Circuit synthesis

Given:

- arbitrary unitary matrix U
- a finite set of gates $\{G_k\}$
- a target precision, ϵ

Find

$$U' = C_1 C_2 \cdots C_m$$

s.t.

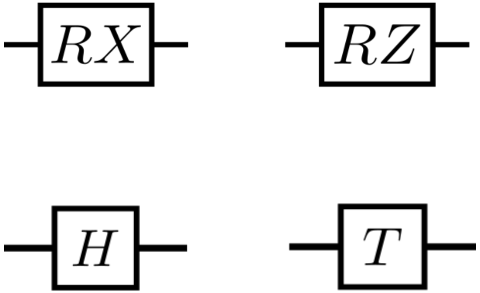
$$||U - U'|| < \epsilon, C_i \in \{G_k\}$$

Universal gate sets

A $\{G_k\}$ for which this can be done is called a **universal gate set**.

Examples for a single qubit:

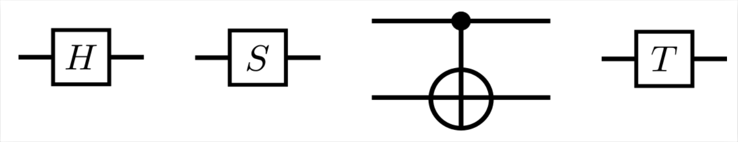
- Any two Pauli rotations (exact)
- Hadamard and *T* (approximate)

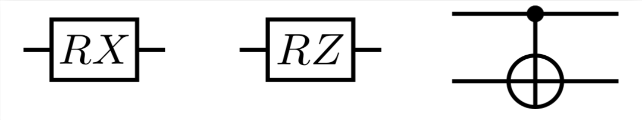


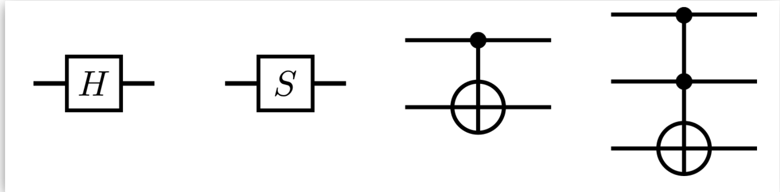
Theoretical bound: $O(\log^c(1/\epsilon))$ gates, *c* varies from 1-4 depending on assumptions.

Universal gate sets

Multi-qubit gate sets







Exercise

Express Pauli X in terms of H and T.

Hint: start by expressing it using H and Z.

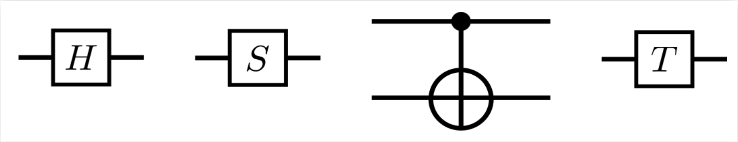
Exercise

Express Hadamard in terms of $\{RX, RY, RZ\}$

Hint: there are many ways to do this!

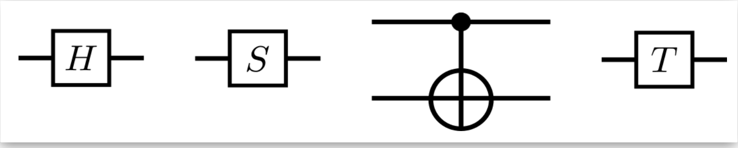
Exercise

Express CZ using gates from



Take-home exercises

1. Express controlled Hadamard using gates from

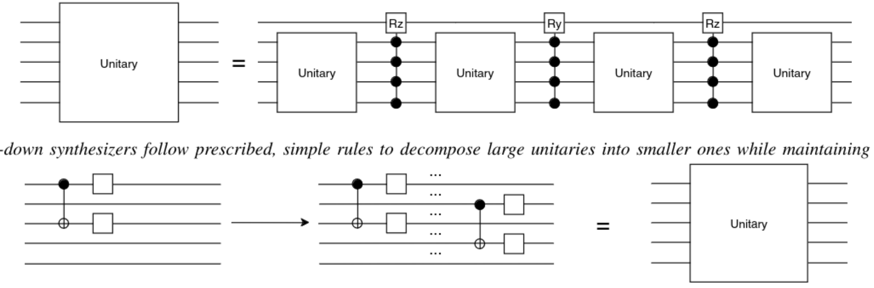


2. Express RZ(π/3) using Hadamard and *T* up to precision 10^{-5}.

Synthesis and decomposition techniques

For under-specified or arbitrary operations, or for special gate sets, we may need to ***find*** a decomposition.

- search problem
- number-theoretic techniques
- numerical methods



(a) *Top-down synthesizers follow prescribed, simple rules to decompose large unitaries into smaller ones while maintaining equality.*

(b) *Bottom-up synthesizers start with an empty circuit and build up to equality.*

Figure 1: *Quantum synthesizers are either top-down or bottom-up.*

Image: E. Younis, K. Sen, K. Yelick, and C. Iancu. *QFAST: Conflating Search and Numerical Optimization for Scalable Quantum Circuit Synthesis.* arXiv:2103.07093 [quant-ph]

Challenges

Synthesis is hard!

- algorithm complexity and computational resources

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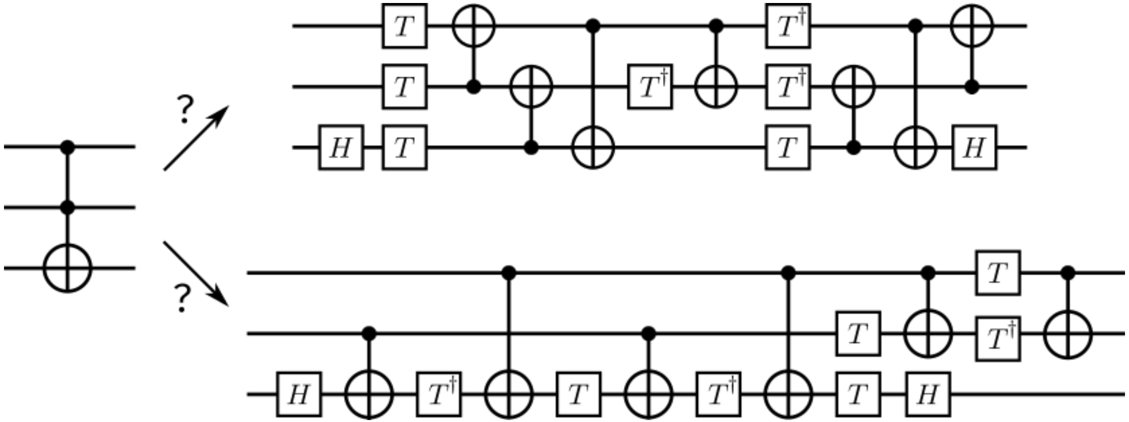
- algorithm complexity and computational resources
- how to choose the best decomposition to *enable further optimization* down the line?



Challenges

Synthesis is hard!

- algorithm complexity and computational resources
- how to choose the best decomposition to *enable further optimization* down the line?
- manage tradeoffs between various resources

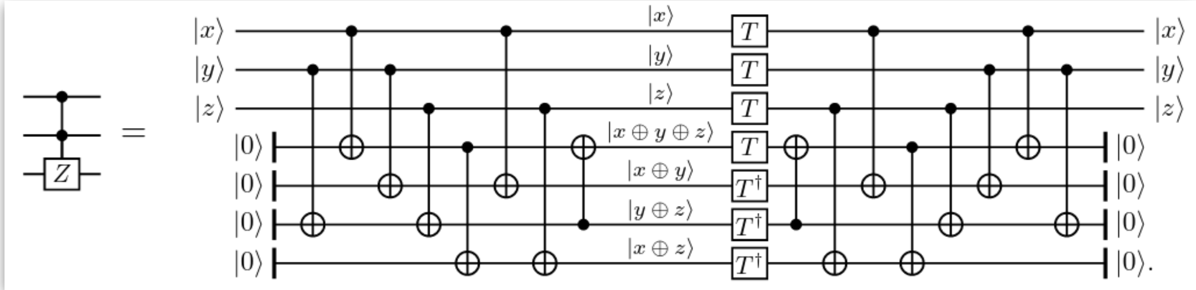
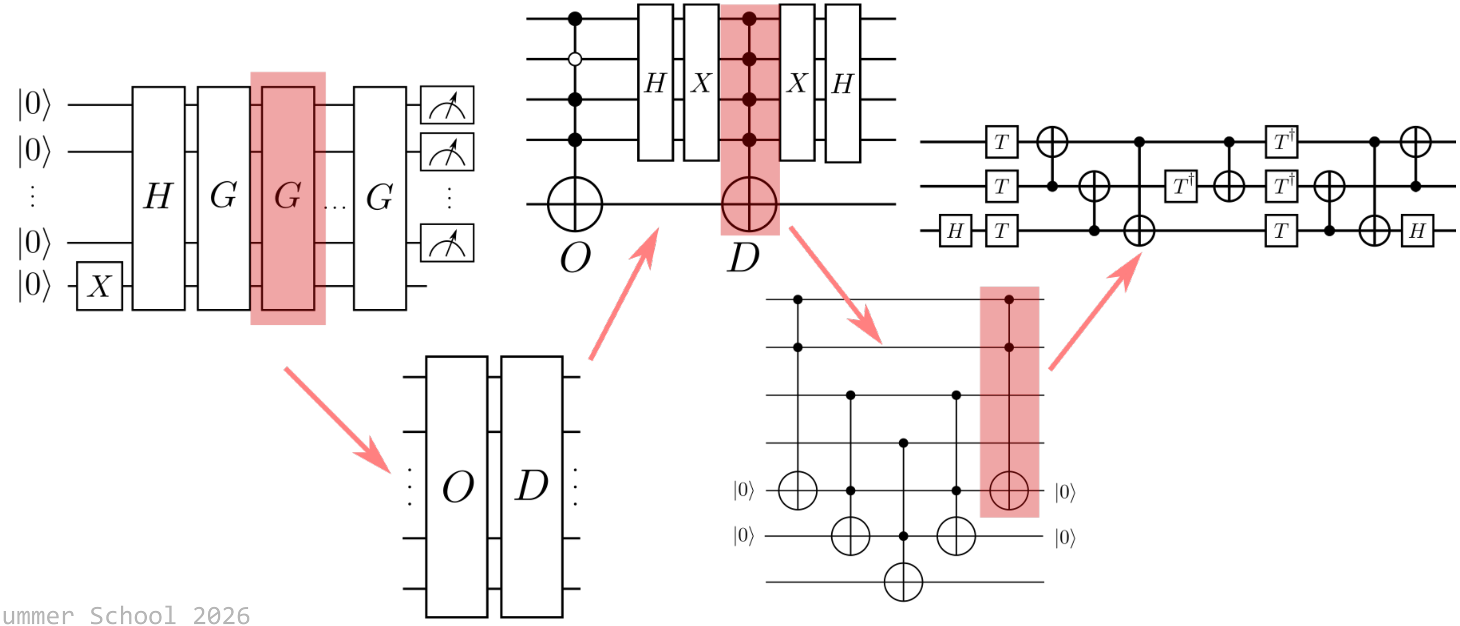


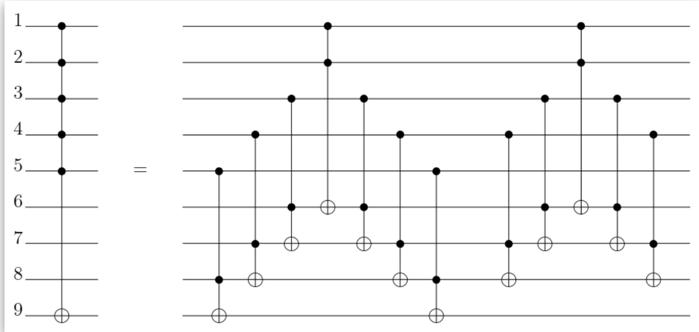
Image: P. Selinger (2013) *Quantum circuits of T -depth one*. Phys. Rev. A 87, 042302 (2013)

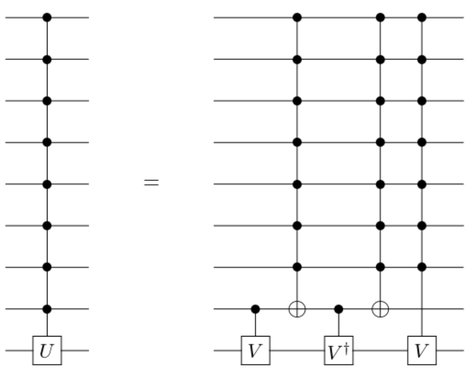
Circuit decomposition

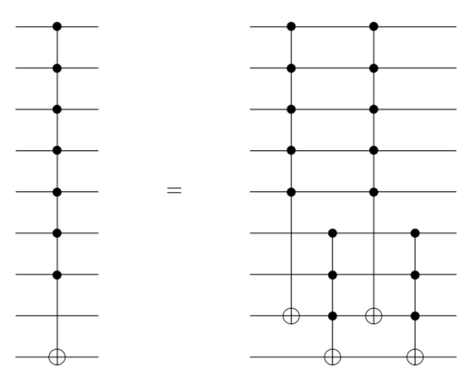
Decompositions of many common operations are **known.** Some are even straightforward to work out by hand.



Multi-controlled NOT: space-time tradeoffs

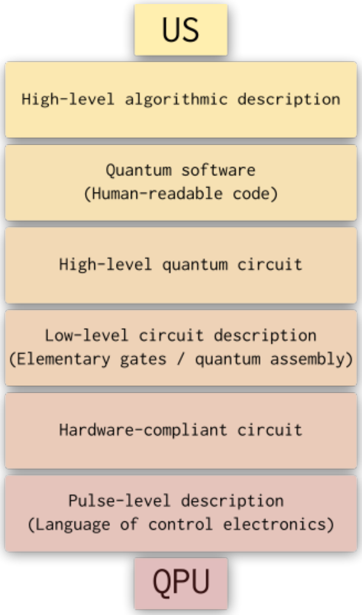






Images: Barenco et al., *Elementary gates for quantum computation* Phys.Rev. A 52 (1995) 3457

Quantum circuit compilation



Implemented as a sequence of modular ***passes***, or ***transforms***, that modify a circuit in various ways.

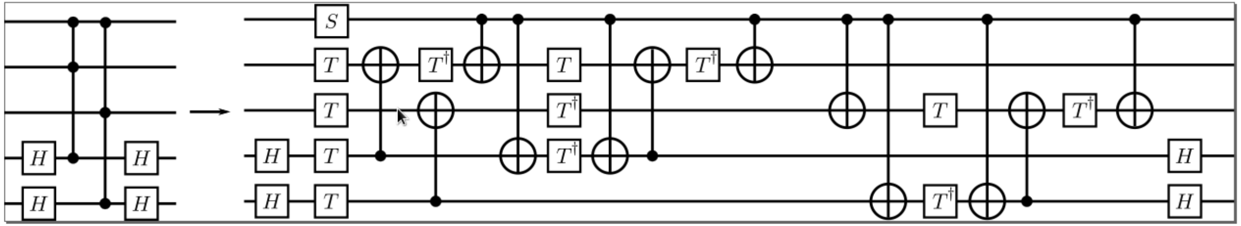
Loosely divided into 3 stages:

- decomposition and synthesis
- hardware-independent optimization
- hardware-dependent optimization

Circuit optimization

Machine-independent (or dependent) sequence of *passes* to reduce:

- gate count (overall, or for particular gate)
- circuit depth
- qubit count



Optimization passes

Passes have varying levels of complexity:

- inverse cancellation

Optimization passes

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- rotation merging

Optimization passes

Passes have varying levels of complexity:

- inverse cancellation
- rotation merging
- template matching

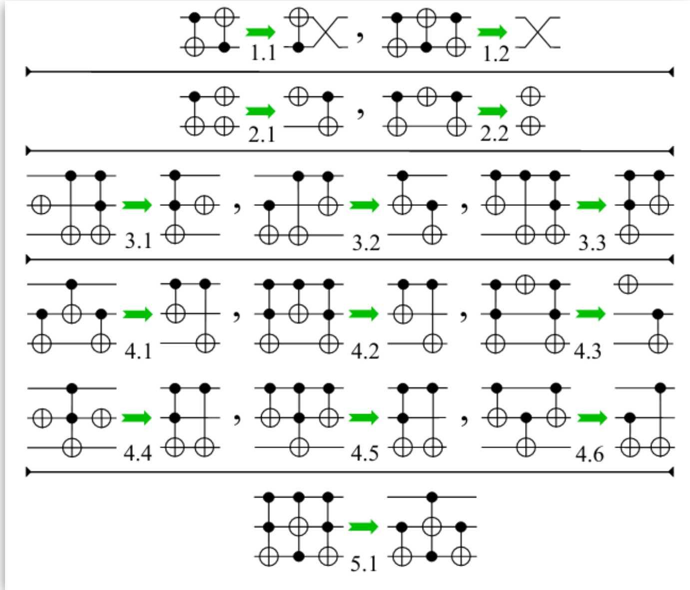
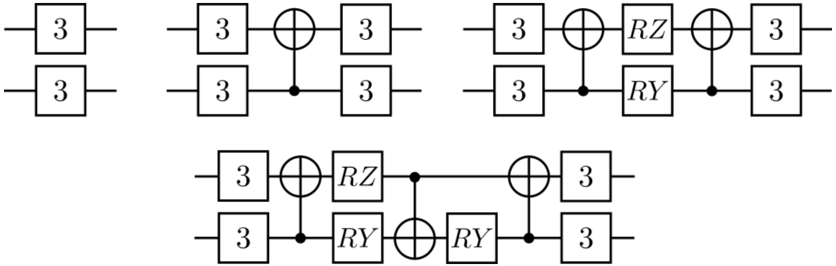


Image: D. M. Miller, D. Maslov, G. W. Dueck. *A Transformation Based Algorithm for Reversible Logic synthesis*. DAC 2003.

Optimization passes

Passes have varying levels of complexity:

- inverse cancellation
- rotation merging
- template matching
- two-qubit block *resynthesis*

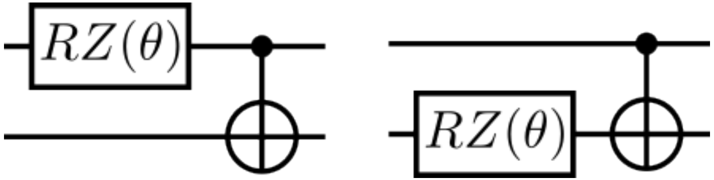


Reference: V. V. Shende, I. L. Markov, and S. S. Bullock (2004) *Minimal universal two-qubit controlled-NOT-based circuits*. Phys. Rev. A 69 062321.

Commutation

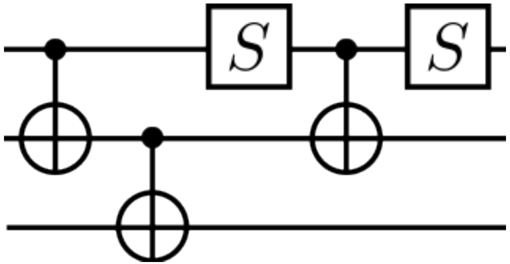
Commutation relations can help us move gates “through” each other to enable optimizations.

Exercise: do the following gates commute?



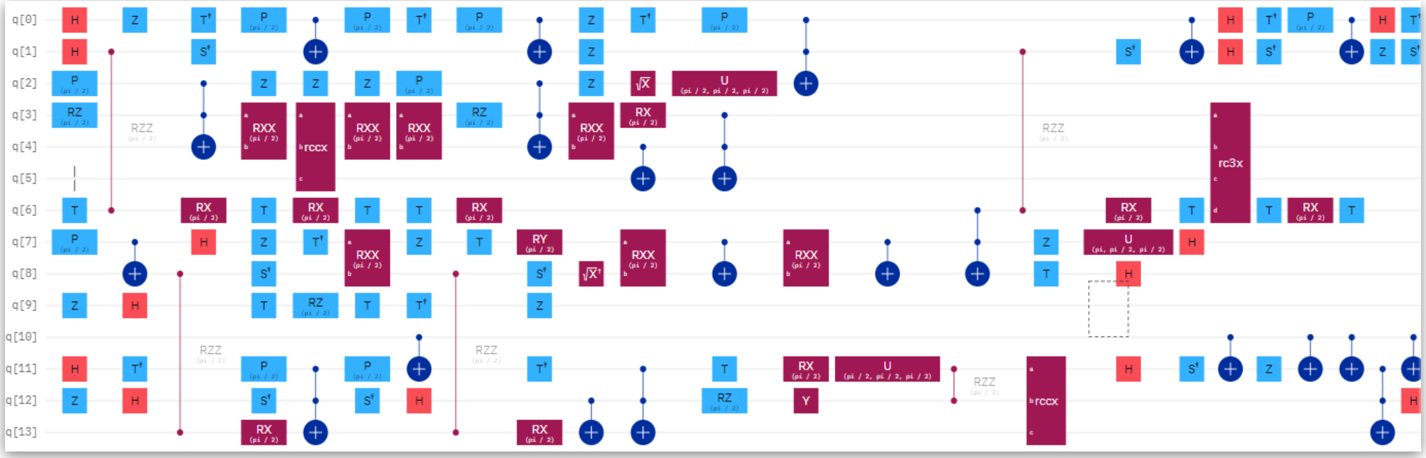
Exercise: easy mode

Determine this circuit’s resource counts, then optimize it.
What is the minimum depth?



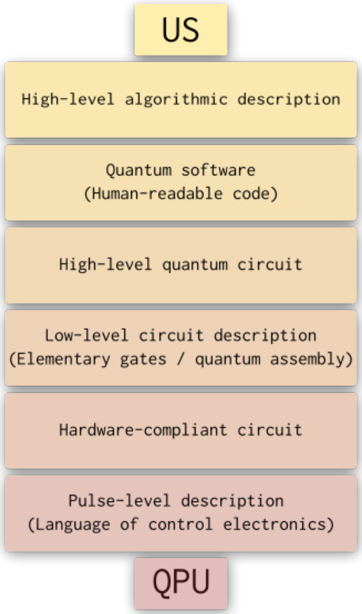
Take-home exercise: hard mode

Determine this circuit's resource counts, then optimize it. What is the minimum depth?



Circuit credit: Mushahid Khan

Quantum circuit compilation



Implemented as a sequence of modular ***passes***, or ***transforms***, that modify a circuit in various ways.

Loosely,

```
new_circuit = compiler_pass(old_circuit)
```

Software for quantum compilation

Most general-purpose QC frameworks include compilation modules.
There are also libraries specializing in compilation.

```python
pass_manager = generate_preset_pass_manager(
    optimization_level=1,
    backend=backend,
    layout_method="trivial",
    routing_method="sabre",
)

compiled_circuit = pass_manager.run(circuit)
```

Qiskit

```python
pipeline = qp.CompilePipeline(
    qp.transforms.commute_controlled,
    qp.transforms.cancel_inverses
    qp.transforms.merge_rotations,
)

compiled_qnode = pipeline(grover_qnode)
```

PENNYLANE

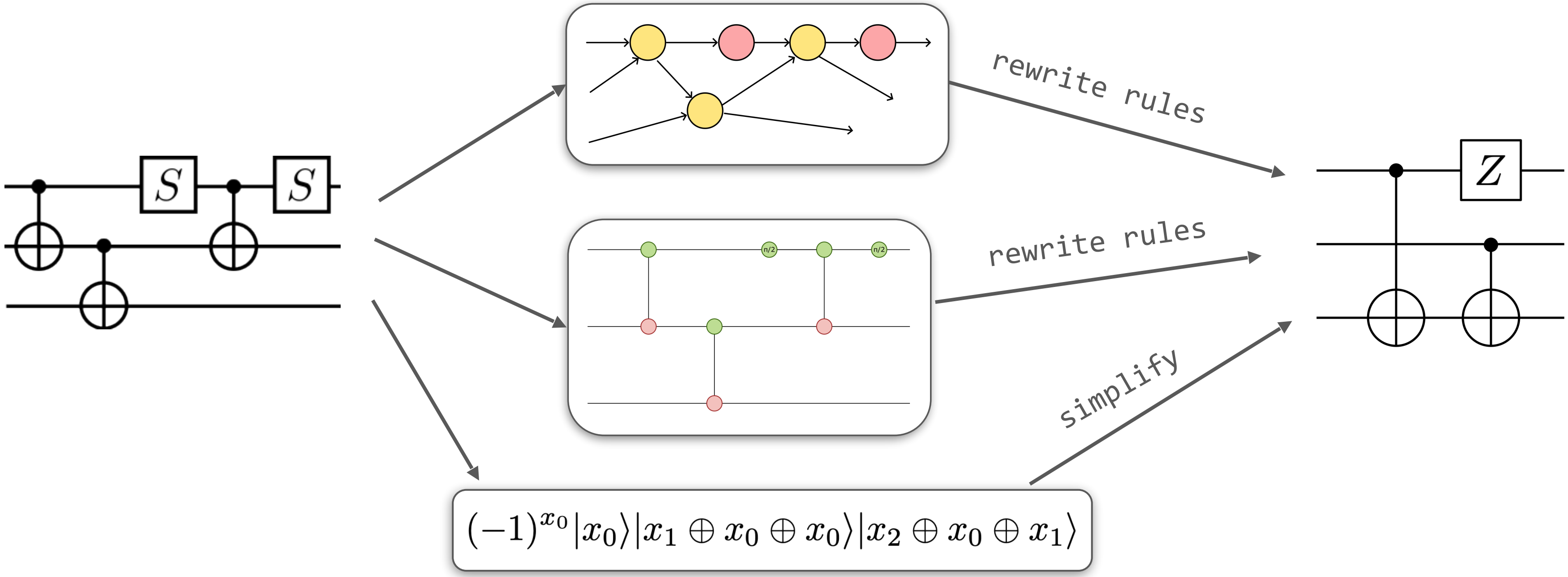
```python
seqpass = SequencePass([
    CommuteThroughMultis(),
    RemoveRedundancies()
])
reppass = RepeatPass(seqpass)

cu = CompilationUnit(circ)
seqpass.apply(cu)
compiled_circuit = cu.circuit
```

TKET

Software for quantum compilation

Pipelines may compose passes acting on different representations.



Software for quantum compilation

Some frameworks contain expansive libraries of passes. Determining a suitable *pass ordering* is an **open problem.**

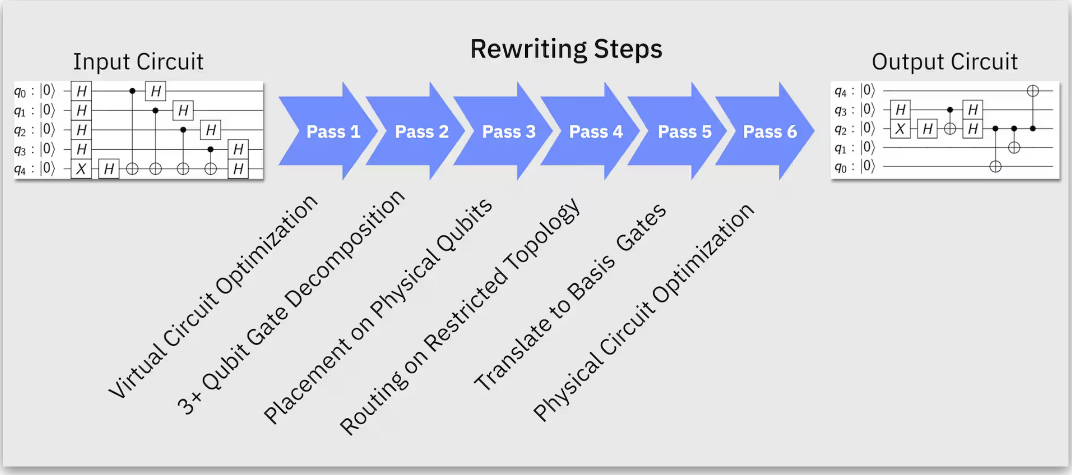
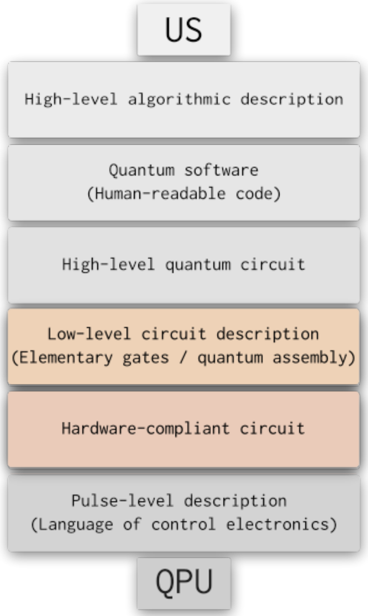


Image: https://quantum.cloud.ibm.com/docs/en/api/qiskit/transpiler#overview

```
passes.RoundAngles()
passes.RoutingPass()
passes.SimplifyInitial()
passes.SimplifyMeasured()
passes.SquashCustom()
passes.SquashRzPhasedX()
passes.SquashTK1()
passes.SynthesiseTK()
passes.SynthesiseTket()
passes.ThreeQubitSquash()
passes.ZXGraphlikeOptimisation()
passes.ZZPhaseToRz()
```

pyTKET passes

Quantum circuit compilation



Implemented as a sequence of modular ***passes***, or ***transforms***, that modify a circuit in various ways.

Loosely divided into 3 stages:

- decomposition and synthesis
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Hardware consideration: gate set

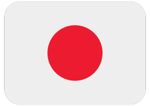
Using a different gate set is like *speaking a different language*.

I would like to use a real quantum computer.

I would like to use a real quantum computer.

J’aimerais utiliser un vrai ordinateur quantique.

I would like to use a real computer that is quantum.

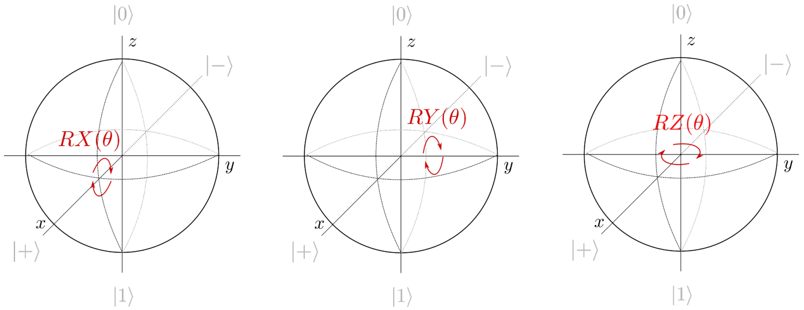


本当の量子コンピュータを使いたい。

A real quantum computer, I would like to use.

Hardware consideration: gate set

RX/RY/RZ are **arbitrary-angle** rotations about **fixed axes**



IonQ's trapped ion hardware native gates: *GPI*/*GPI2*, **fixed-angle** rotations about **arbitrary axes** in *xy*-plane

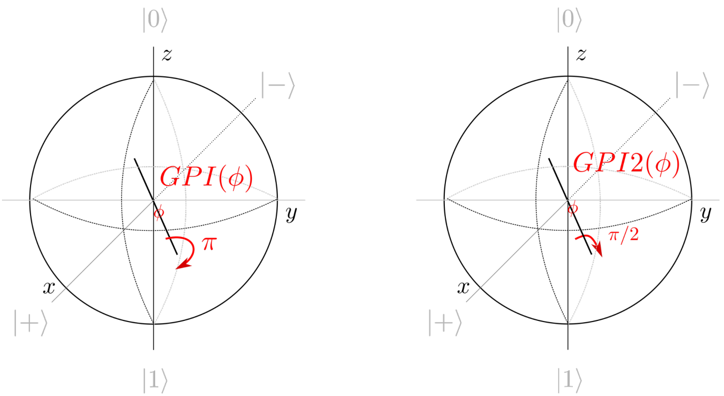
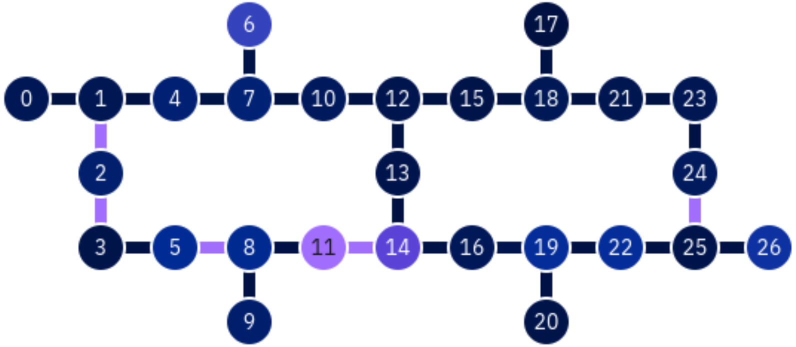


Image adapted from Xanadu Quantum Codebook Node I.6
See blog post: https://pennylane.ai/blog/2023/06/the-ionizer-building-a-hardware-specific-transpiler-using-pennylane

Qubit placement and routing

Each type of machine has its own compilation challenges:

- superconducting machines have restricted topology



IBM Q Kolkata – screenshot 2023-08-24

Qubit placement and routing

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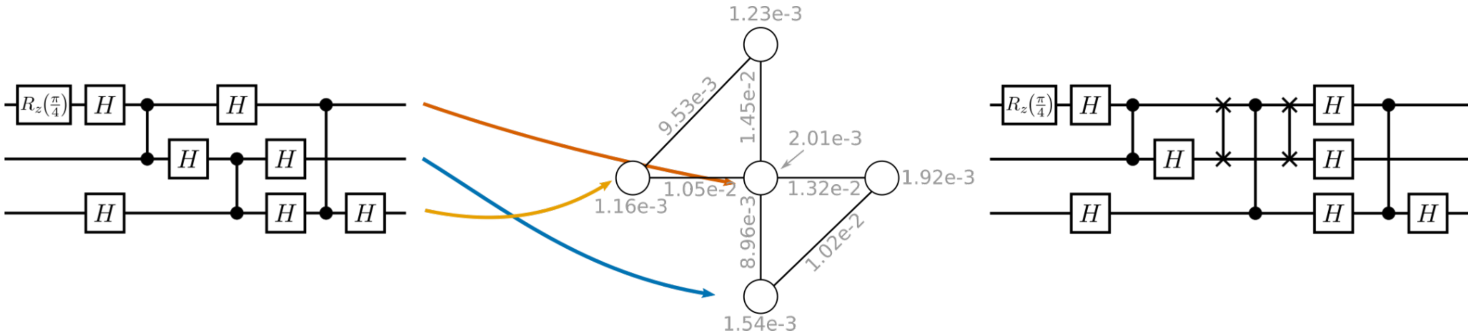
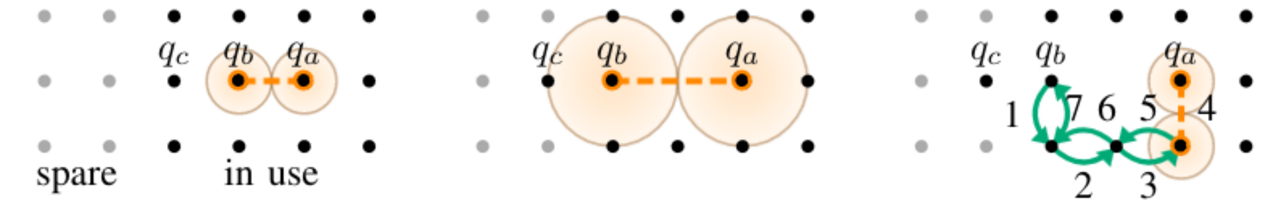


Placement is NP-complete; routing is NP hard.

Qubit placement and routing

Each type of machine has its own compilation challenges:

- superconducting machines have restricted topology
- neutral atom machines can lose atoms



(a) Circuit Before Loss (b) Virtual Remapping (c) Reroute

Image: Baker et al. (2021) *Exploiting Long-Distance Interactions and Tolerating Atom Loss in Neutral Atom Quantum Architectures.* ISCA ‘21.

Qubit placement and routing

Techniques can use optimal solvers, heuristics, or a combination of both.

Example: SABRE (“SWAP-based BidiREctional heuristic search)

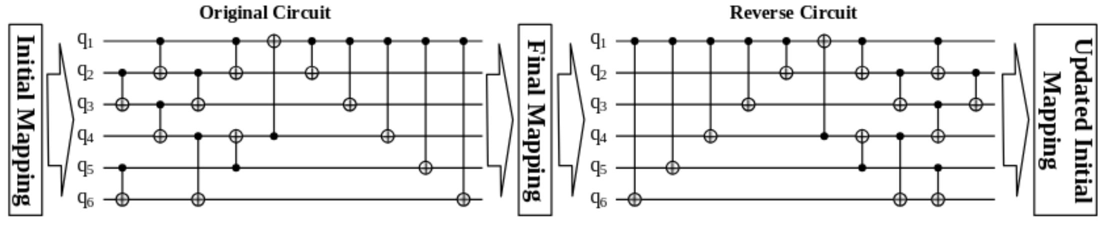


Fig. 5: Initial Mapping Update Using Reverse Traversal Technique

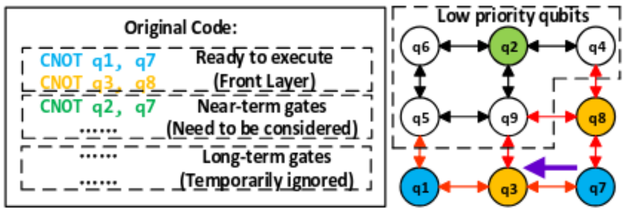
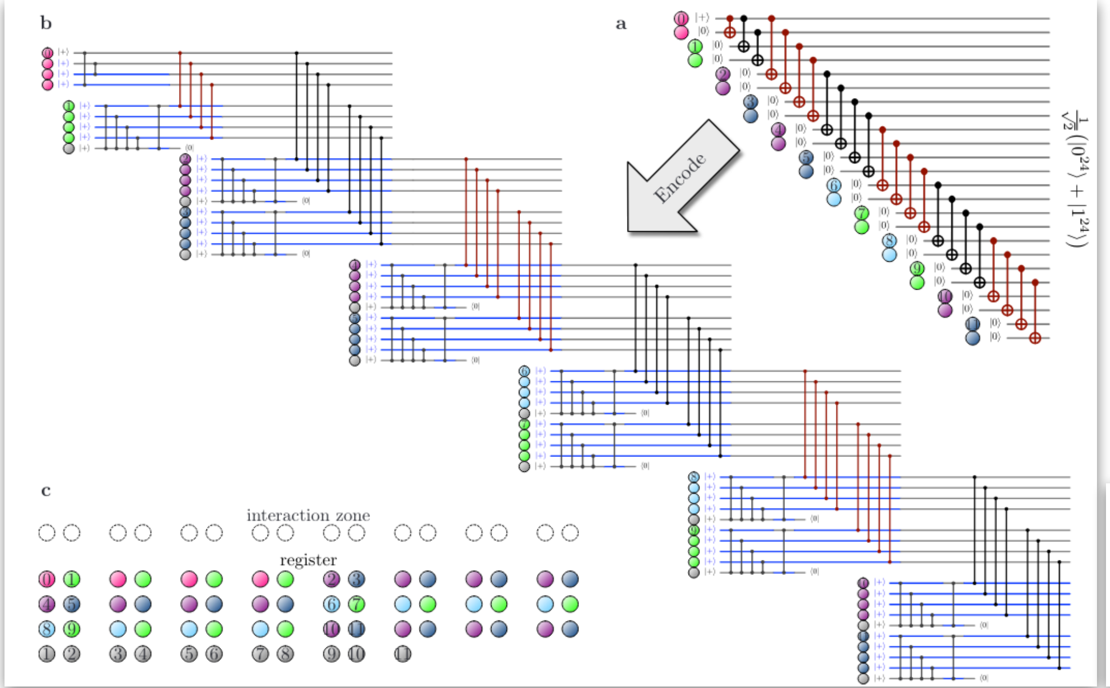


Fig. 6: Example of SWAP-Based Heuristic Search

G. Li, Y. Ding, and Y. Xie. 2019. Tackling the Qubit Mapping Problem for NISQ-Era Quantum Devices. In Proceedings of the Twenty-Fourth International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '19). ACM, New York, NY, USA, 1001–1014.

Gate scheduling



Recent demonstration of logical qubits on neutral atom machine: *move qubits in and out of "interaction zone" for 2-qubit gates*

(Microsoft + Atom computing + collaborators)

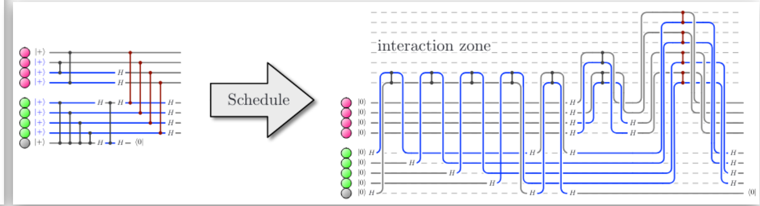
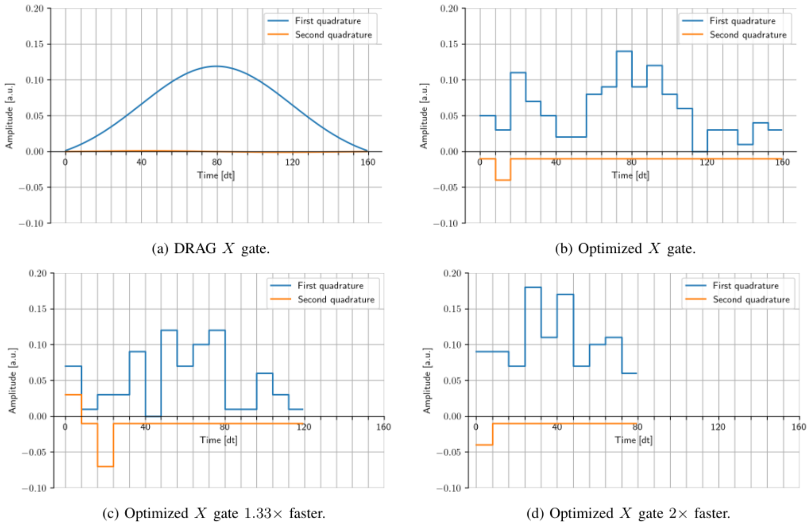


Image: Fig. 4 and 5, B. W. Reichardt et al., "Logical computation demonstrated with a neutral atom quantum processor". arXiv:2411.11822 [quant-ph].

Pulse design and optimization

Electromagnetic pulses *actually* implement the gates. Interesting problems:

- optimizing duration/shape



(a) DRAG X gate.

(b) Optimized X gate.

(c) Optimized X gate 1.33× faster.

(d) Optimized X gate 2× faster.

E. Wright and R. De Sousa, "Fast Quantum Gate Design with Deep Reinforcement Learning Using Real-Time Feedback on Readout Signals," in 2023 IEEE International Conference on Quantum Computing and Engineering (QCE), Bellevue, WA, USA, 2023, pp. 1295-1303

Pulse design and optimization

Electromagnetic pulses *actually* implement the gates. Interesting problems:

- optimizing duration/shape
- scheduling constraints

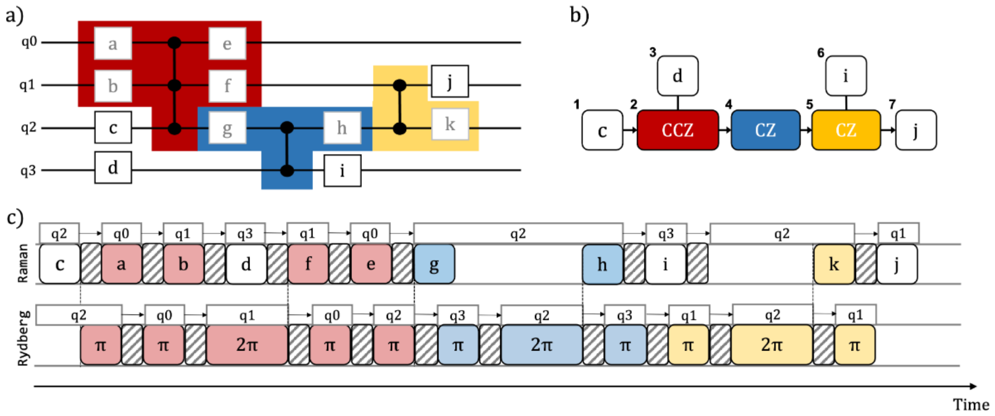


Image: R. B.-S. Tsai, H. Silvério, L. Henriet (2022) *Pulse-Level Scheduling of Quantum Circuits for Neutral-Atom Devices*. Phys. Rev. Applied 18 064035.

Big problem: complexity and scale

Many tasks still require **human labour** and **expert experience**.

- How to better **automate** the process?
- How to **scale** to multi-QPU systems with 1000s of qubits?
- How to **verify** compiler output for circuits too large to simulate?

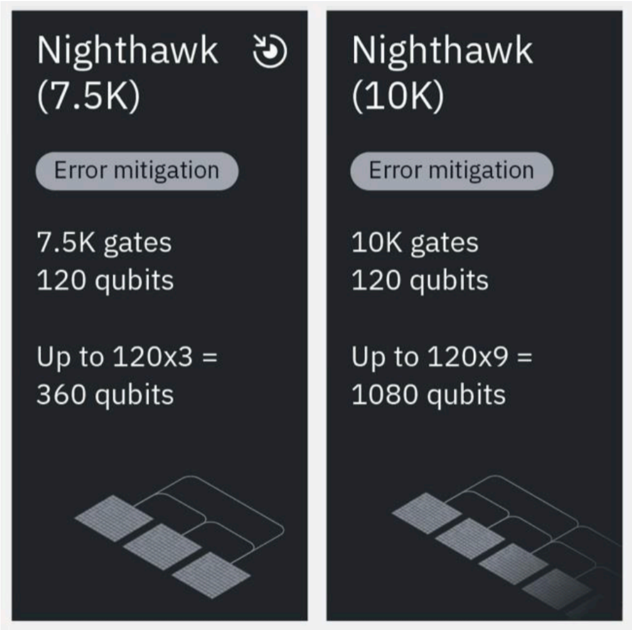


Image: IBM Quantum Roadmap (https://www.ibm.com/downloads/documents/us-en/1443d5cda24021e4)

Bigger problem: error correction

We must revisit our assumptions and make some new ones…

- How good is the hardware?
- What code parameters do we need?
- What is the overhead of encoding?
- How do we apply logical operations?
- Which gates are expensive?